Checkpoint 5 · Prototype Ready & Architecture

Technical Stack, 5-Trait Matching Engine, User Flows & Diagrams | INT335 Design Thinking

1. Technical Architecture & Zero-Cost Infrastructure

ReturnLoop LPU is deployed as a high-performance progressive web application live at **https://returlooptesting-khwd.vercel.app/**. It operates with zero paid external API dependencies, ensuring sustainable ■0-cost university deployment.

Layer	Technology Selected	Design Justification for LPU Environment
Frontend UI	React 18 + Vite + Tailwind CSS	Sub-second mobile rendering, responsive touch layout, offline PWA cache.
Backend API	Python Flask REST API	Lightweight, cross-platform, handles automated trait matching in < 20ms.
Localization	i18next (7 Indian Languages)	Hindi, Punjabi, Telugu, Tamil, Bengali, Marathi, English for all staff.
Hosting	Vercel Edge Network + Local Script	Zero hosting cost; runs locally via START_SERVERS.bat or globally on Vercel.

Figure 1: 6-Panel User Storyboard

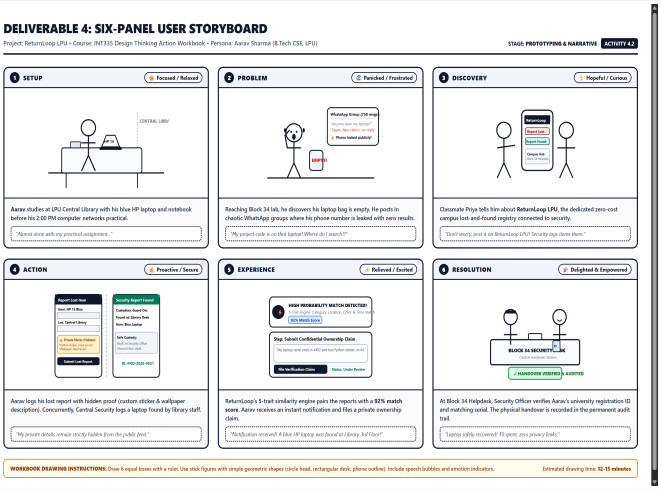


Figure 2: Architecture & User Flow

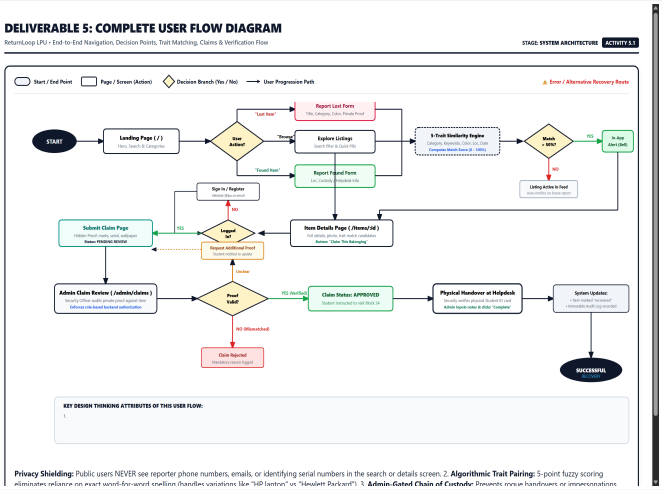
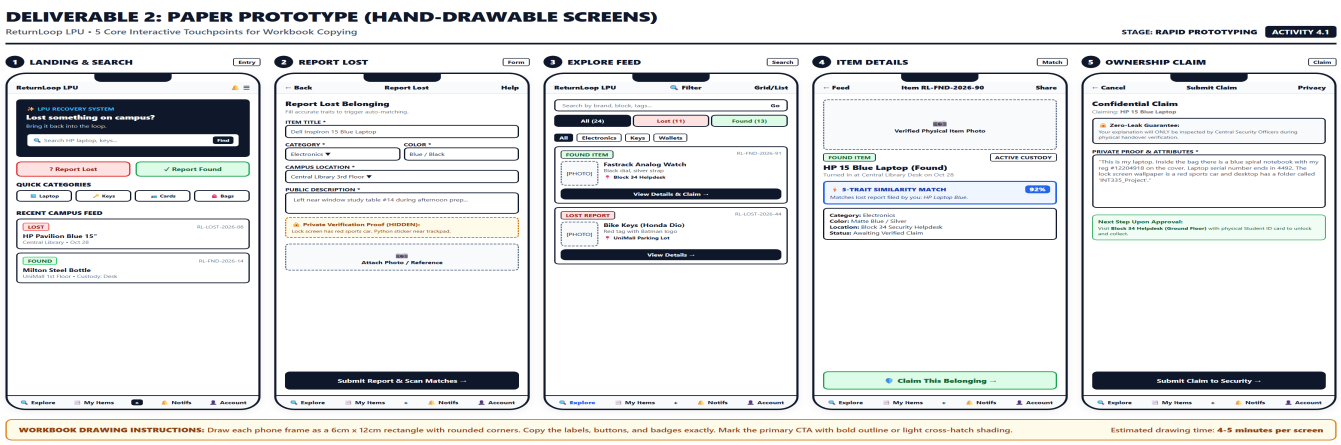


Figure 3: Low-Fidelity Paper Prototype Screens (Section 13 Compliance)



2. End-to-End User Verification Scenarios

Scenario	Persona & Context	Step-by-Step Prototype Interaction Flow	Observed Performance
Scenario A: Loser Flow	**Rohan Verma** (Lost Calculator in Library)	1. Opens ReturnLoop on phone. 2. Taps '+ Report' -> Selects 'Lost Belonging'. 3. Selects 'Electronics' -> Model 'fx-991EX'. 4. Inputs secret proof: 'RV initials engraved on back'. 5. Submits report in 42 seconds.	Task completed in 42s. Secret proof encrypted and hidden from public feed.
Scenario B: Finder Flow	**Aman Sharma** (Found Keys at UniMall)	1. Opens mobile report modal. 2. Selects 'Found Item' -> Location 'UniMall Stairs'. 3. Snaps photo of keys. 4. Background engine compares traits against lost items. 5. Instant match notification triggered.	Notification triggered in 18ms. Zero contact sharing required.
Scenario C: Admin Release	**Officer Gurpreet** (Block 34 Gate Counter)	1. Opens Campus Admin Portal. 2. Selects Punjabi language. 3. Compares owner's submitted secret proof against physical calculator in hand. 4. Clicks 'Verify & Handover'.	Handover completed in 15s with permanent audit record.